

# Vaibhav Lalwani

AI & Full-Stack Engineer · MSc Advanced Data Science & AI, University of Liverpool

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Open to: Full-time Part-time Contract Remote Hybrid On-site

## PROFILE

AI engineer building production language-model systems end-to-end — inference, retrieval, evaluation harnesses and the front ends that ship them. Comfortable across the model–system boundary. Author of an open-access preprint on on-controller transformer inference (NEXUS, 2026). Two years freelancing 20+ LLM-powered apps for paying clients on Claude, OpenAI and LangChain.

## PUBLICATION

**Lalwani, V.** (2026). *NEXUS: On-Controller Transformer Inference and Speculative Edge Execution for Console-Wired Latency Parity in Cloud Gaming*. Zenodo. DOI [10.5281/zenodo.20059414](https://doi.org/10.5281/zenodo.20059414) · [ResearchGate](https://ResearchGate). Extends the Outatime line (MobiSys 2015) with on-controller transformer inference and speculative edge execution.

## SKILLS

LANGUAGES Python · TypeScript · JavaScript (ES6+) · SQL · HTML · CSS  
AI / ML OpenAI · Anthropic / Claude · Gemini · Groq · LangChain · LangSmith · RAG · structured outputs · MCP servers · Whisper STT · PyTorch · transformer inference  
BACKEND FastAPI · Node.js / Express · Postgres · pgvector · Redis · BullMQ · Supabase (RLS) · vector search  
FRONTEND React 19 · Next.js 15 / 16 · Tailwind CSS · Vite · REST + GraphQL · accessibility  
DEVOPS Docker · GitHub Actions · Vercel · GCP Cloud Run · AWS · Sentry · PostHog · Jest / Vitest · Playwright · Puppeteer

## EXPERIENCE

**Full-Stack / AI Engineer Intern · Meta Solution Technologies** Apr 2026 – Present  
*Remote, UK*

- Building an AI-guided admissions platform on Next.js 15 + React 19 + TypeScript. Chat-style guidance flow on OpenAI with structured outputs, served from a Python FastAPI service against Postgres + pgvector.
- Wrote auth end-to-end (Google OAuth, magic-link, PKCE) on Supabase, plus a BullMQ + Redis async queue driving document checks, generation jobs and email triggers.
- Shipped a 10-language i18n layer and a dual-theme CSS system so non-engineering teammates roll out copy and translation updates without engineering involvement.

**AI Engineer · Freelance / Self-employed** May 2024 – Jan 2026  
*Remote*

- Designed and shipped 20+ LLM-powered applications for paying clients across SaaS, e-commerce and consulting. Most pair a React or Next.js front end with a Python service running Claude or OpenAI APIs, structured outputs and a RAG retrieval layer.
- Built longer-running AI agents for lead generation, customer support and data processing. Each project shipped with a README, a walkthrough video and a written hand-off doc.

**Software Engineer Intern · Recruit Pilot** Jan 2026 – Apr 2026  
*Recruitment Technology · Remote*

- Built React + TypeScript UI features against a REST API. Code review and weekly sprint cadence alongside more senior engineers.

## SELECTED PROJECTS

**apex** — autonomous job application engine. CLI driving real Chrome via Playwright, generates a tailored 1-page resume per job (Puppeteer markdown → ATS-safe PDF), fills custom LinkedIn Easy Apply questions with an LLM (profile mapping → Q+A cache → free-LLM fallback), submits autonomously until LinkedIn's daily cap. Free LLMs only. [github.com/vaibhav4046/apex](https://github.com/vaibhav4046/apex)

**Praxon** — open-source AI agent platform. A Claude Cowork alternative on Next.js 16 + React 19 + TypeScript. Multi-LLM router across free providers (Groq, Cerebras, Gemini, Ollama) with auto-fallback; MCP-native tool layer; 3-schema Postgres + RLS for tenant isolation; cloud-deployable on Vercel + Supabase. [Live](#) · [Code](#)

**Cogniloop** — Socratic study tool. Locked-prompt evaluator grading free-form student answers 0–3 with explanation; tracks concept mastery (weak → mastered) across sessions. Single-prompt design — no agent loop, no retrieval — keeps median latency under one second on Groq Llama 3.3 70B. Edge runtime, Next.js 16, KaTeX. [Live](#) · [Code](#)

**MCP Marketplace** — registry of 800+ Model Context Protocol servers. Daily auto-sync from Glama and the official MCP repo, normalised tool schemas, one-line install snippets for Claude Desktop, Cursor and Claude Code. Next.js 15 RSC, Cmd-K palette, dynamic OG cards. [Live](#) · [Code](#)

## EDUCATION

**University of Liverpool** Jan 2026 – Jan 2027  
*MSc Advanced Data Science & Artificial Intelligence*

**Christ University, Bengaluru** 2022 – 2025  
*Bachelor of Computer Applications · CGPA 7.86 / 10*